# The Association Between Platelet Indices and Presence and Severity of Psoriasis – A Systematic Review and Meta-Analysis

Clinical and Experimental Medicine

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# Supplementary Information

## Table of Contents


*Online Resource 1: Search Strategy* ..... *3*

**MEDLINE (Ovid)** ..... **3**

**Embase (Ovid)** ..... **3**

**Cochrane Library** ..... **3**

*Online Resource 2: Newcastle-Ottawa Scale (NOS) Assessment of Methodological Quality* ..... *4*

*Leave-one-out Sensitivity Analyses* ..... *5*

**Online Resource 3** ..... **5**

**Online Resource 6** ..... **6**

**Online Resource 7** ..... **7**

**Online Resource 10** ..... **8**

**Online Resource 13** ..... **9**

**Online Resource 14** ..... **9**

*Meta-regression Tables* ..... *10*

**Online Resource 4: Meta-regression results of PLT and psoriasis** ..... **10**

**Online Resource 8: Meta-regression results of MPV and psoriasis** ..... **11**

**Online Resource 11: Meta-regression results of MPV and PASI** ..... **12**

*Supplemental Figures* ..... *13*

**Online Resource 5** ..... **13**

**Online Resource 9** ..... **14**

**Online Resource 12** ..... **15**

# Online Resource 1: Search Strategy

## MEDLINE (Ovid)

1. exp Mean Platelet Volume/ or mean platelet volume.mp. or MPV.mp.
2. exp Platelet Count/ or (platelet$ or platelet count$ or platelet number$).mp.
3. (platelet distribution width$ or PDW).mp.
4. (plateletcrit$ or PCT).mp.
5. exp Cell-Derived Microparticles/ or Platelet-derived microparticle$.mp.
6. (Immature platelet fraction$ or IPF).mp.
7. (platelet mass ind$ or PMI).mp.
8. psoriasis.mp. or exp Psoriasis/
9. 1 or 2 or 3 or 4 or 5 or 6 or 7
10. 8 and 9

## Embase (Ovid)

1. exp Mean Platelet Volume/ or mean platelet volume.mp. or MPV.mp.
2. exp Platelet Count/ or (platelet$ or platelet count$ or platelet number$).mp.
3. (platelet distribution width$ or PDW).mp.
4. (plateletcrit$ or PCT).mp.
5. Platelet-derived microparticle$.mp. or exp platelet microparticle/
6. (Immature platelet fraction$ or IPF).mp.
7. (platelet mass ind$ or PMI).mp.
8. psoriasis.mp. or exp Psoriasis/
9. 1 or 2 or 3 or 4 or 5 or 6 or 7
10. 8 and 9
11. 10 not Conference Abstract.pt.

## Cochrane Library

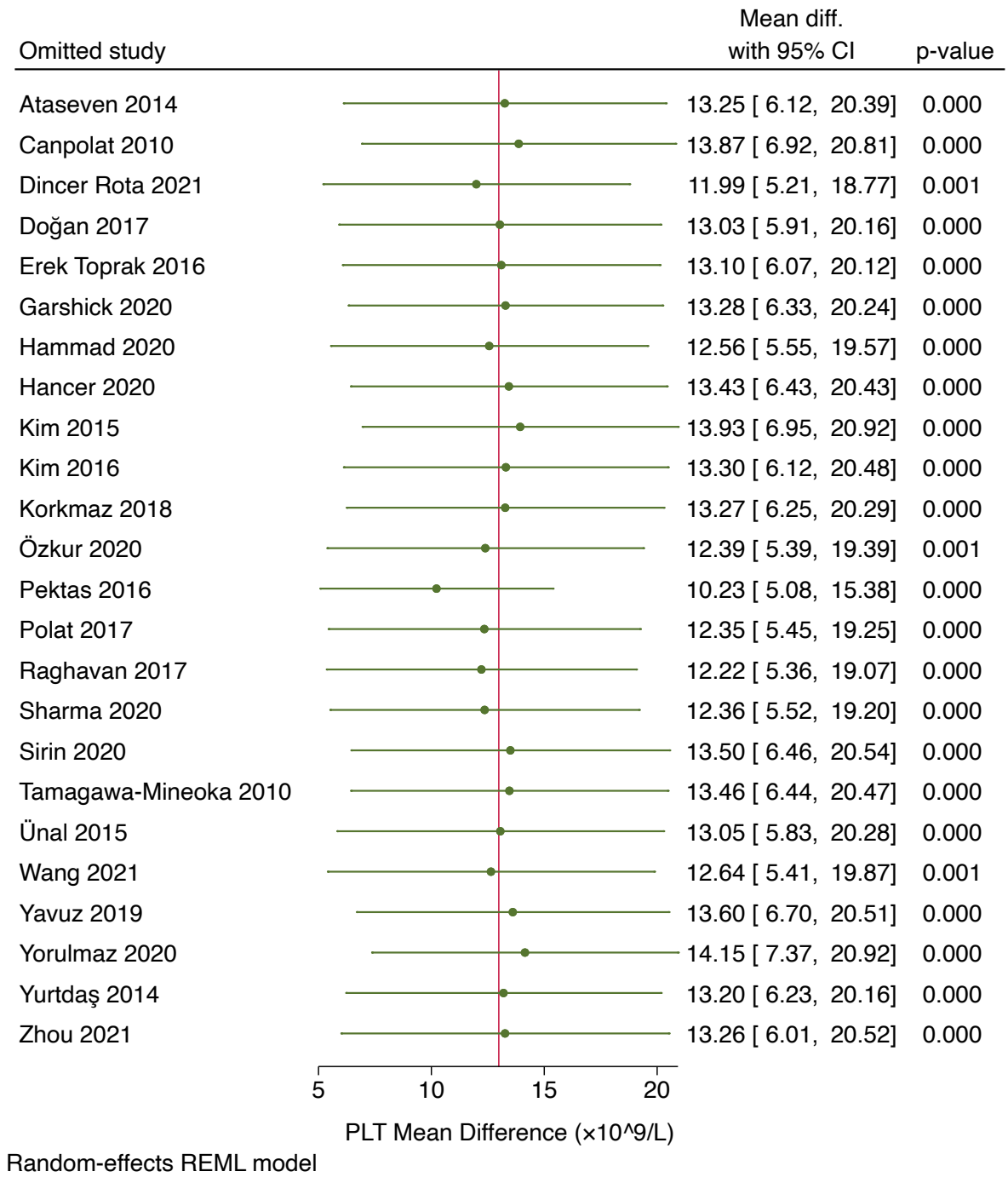
| | |
|---|---|
| #1 | #1 MeSH descriptor: [Mean Platelet Volume] explode all trees |
| #2 | mean platelet volume |
| #3 | MPV |
| #4 | MeSH descriptor: [Platelet Count] explode all trees |
| #5 | platelet* or platelet count* or platelet number* |
| #6 | platelet distribution width* or plateletcrit* |
| #7 | PDW or PCT |
| #8 | Platelet derived microparticle* |
| #9 | Immature platelet fraction* |
| #10 | IPF |
| #11 | Platelet mass ind* |
| #12 | PMI |
| #13 | #1 or #2 or #3 or #4 or #5 or #6 or #7 or #8 or #9 or #10 or #11 or #12 |
| #14 | Psoriasis |
| #15 | MeSH descriptor: [Psoriasis] explode all trees |
| #16 | #14 or #15 |
| #17 | #13 and #16 |

# Online Resource 2: Newcastle-Ottawa Scale (NOS) Assessment of Methodological Quality

| Newcastle-Ottawa Scale for Case-Control Studies | OVERALL ASSESSMENT | | Selection | | | | Comparability | Exposure | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Study | | | Is the case definition adequate? | Representativeness of the cases | Selection of controls | Definition of controls | Comparability of cases and controls on the basis of the design or analysis | Ascertainment of exposure | Same method of ascertainment for cases and controls | Non-response rate |
| | We considered astudy which scored ≥ 7 as high-quality, ≥ 5 as fair quality, and the remaining as poor quality. | Star Count | a) Requires some independent validation (e.g. >1 person/record/time/process to extract information, or reference to primary record source such as x-rays or medical/hospital records)* b) Record linkage (e.g. ICD codes in database) or self-report with no reference to primary record c) No description | a) All eligible cases with outcome of interest over a defined period of time, all cases in a defined catchment area, all cases in a defined hospital or clinic, group of hospitals, health maintenance organisation, or an appropriate sample of those cases (e.g. random sample)* b) Not satisfying requirements in part (a), or not stated. | This item assesses whether the control series used in the study is derived from the same population as the cases and essentially would have been cases had the outcome been present. a) Community controls (i.e. same community as cases and would be cases if had outcome)* b) Hospital (including non-cosmetic outpatient dermatology) controls, within same community as cases (i.e. not another city) but derived from a hospitalised (including non-cosmetic outpatient dermatology) population c) No description | a) If cases are first occurrence of outcome, then it must explicitly state that controls have no history of this outcome. If cases have new (not necessarily first) occurrence of outcome, then controls with previous occurrences of outcome of interest should not be excluded.* b) No mention of history of outcome | A maximum of 2 stars can be allotted in this category Either cases and controls must be matched in the design and/or confounders must be adjusted for in the analysis. Statements of no differences between groups or that differences were not statistically significant are not sufficient for establishing comparability. Note: If the odds ratio for the exposure of interest is adjusted for the confounders listed, then the groups will be considered to be comparable on each variable used in the adjustment. There may be multiple ratings for this item for different categories of exposure (e.g. ever vs. never, current vs. previous or never) a) study controls for age * b) study controls for gender * | a) secure record (eg surgical records) * b) structured interview where blind to case/control status * c) interview not blinded to case/control status d) written self report or medical record only e) no description | a) Yes* b) No | a) Same for both groups* b) Non-respondents described c) Rate different and no designation |
| Ahmad 2014 | High | 7 | 0 | 1 | 0 | 1 | 2 | 1 | 1 | 1 |
| Ataseven 2014 | Fair | 5 | 1 | 0 | 0 | 1 | 0 | 1 | 1 | 1 |
| Canpolat 2010 | Fair | 6 | 0 | 0 | 0 | 1 | 2 | 1 | 1 | 1 |
| Çerman 2016 | Fair | 5 | 1 | 0 | 0 | 1 | 0 | 1 | 1 | 1 |
| Chandrashekar 2015 | High | 7 | 0 | 0 | 1 | 1 | 2 | 1 | 1 | 1 |
| Dincer Rota 2021 | High | 7 | 1 | 0 | 0 | 1 | 2 | 1 | 1 | 1 |
| Doğan 2017 | High | 7 | 1 | 0 | 0 | 1 | 2 | 1 | 1 | 1 |
| Erek Toprak 2016 | Poor | 4 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 |
| Farag 2018 | High | 7 | 0 | 0 | 1 | 1 | 2 | 1 | 1 | 1 |
| Garshick 2020 | High | 9 | 1 | 1 | 1 | 1 | 2 | 1 | 1 | 1 |
| Hammad 2020 | Fair | 6 | 0 | 0 | 0 | 1 | 2 | 1 | 1 | 1 |
| Hancer 2020 | Fair | 6 | 0 | 0 | 0 | 1 | 2 | 1 | 1 | 1 |
| Işik 2016 | Poor | 4 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 |
| Karabudak 2008 | Fair | 6 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 |
| Kim 2015 | Fair | 5 | 1 | 0 | 0 | 1 | 0 | 1 | 1 | 1 |
| Kim 2016 | High | 8 | 1 | 0 | 1 | 1 | 2 | 1 | 1 | 1 |
| Kılıç 2017 | Fair | 6 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 1 |
| Korkmaz 2018 | Fair | 6 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 1 |
| Mahrous 2018 | Fair | 5 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 |
| Özkur 2020 | High | 7 | 0 | 0 | 1 | 1 | 2 | 1 | 1 | 1 |
| Pektas 2016 | Fair | 5 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 1 |
| Polat 2017 | High | 8 | 1 | 0 | 1 | 1 | 2 | 1 | 1 | 1 |
| Raghavan 2017 | Fair | 5 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 |
| Saleh 2013 | High | 7 | 0 | 0 | 1 | 1 | 2 | 1 | 1 | 1 |
| Sharma 2020 | Fair | 5 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 |
| Sirin 2020 | Fair | 6 | 0 | 0 | 0 | 1 | 2 | 1 | 1 | 1 |
| Tamagawa-Mineoka 2010 | Fair | 6 | 0 | 0 | 0 | 1 | 2 | 1 | 1 | 1 |
| Ünal 2016 | Fair | 5 | 1 | 0 | 0 | 1 | 0 | 1 | 1 | 1 |
| Wang 2021 | High | 8 | 1 | 0 | 1 | 1 | 2 | 1 | 1 | 1 |
| Yavuz 2019 | Fair | 5 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 1 |
| Yorulmaz 2020 | High | 8 | 1 | 0 | 1 | 1 | 2 | 1 | 1 | 1 |
| Yurtdaş 2014 | High | 7 | 0 | 1 | 0 | 1 | 2 | 1 | 1 | 1 |
| Zhou 2021 | Fair | 6 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 1 |

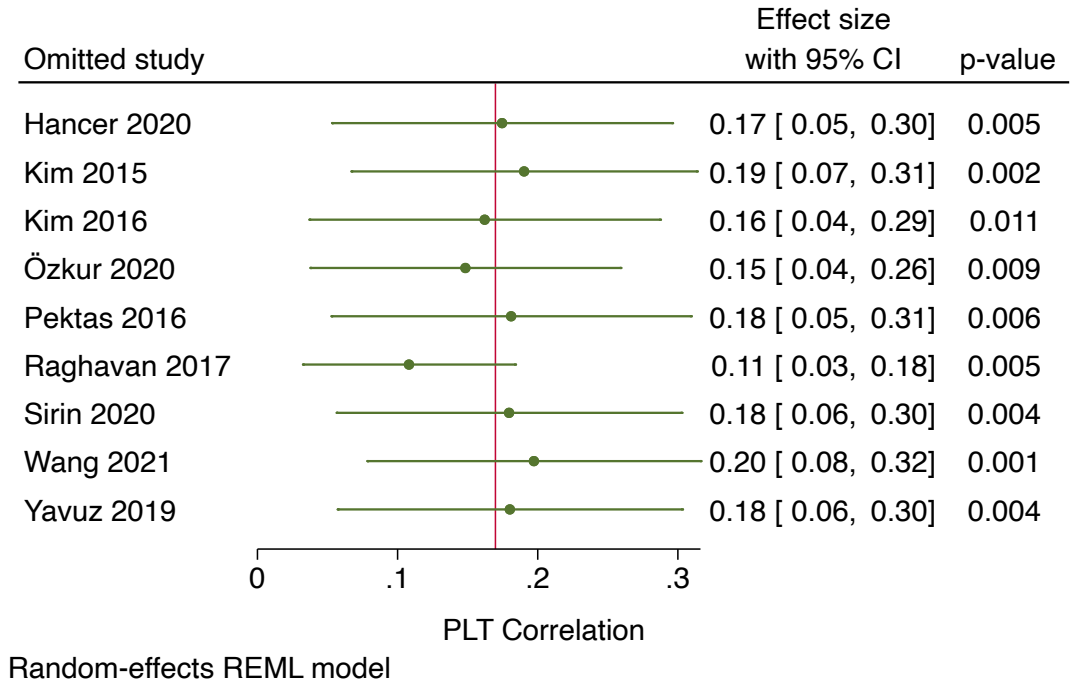
# Leave-one-out Sensitivity Analyses

## Online Resource 3



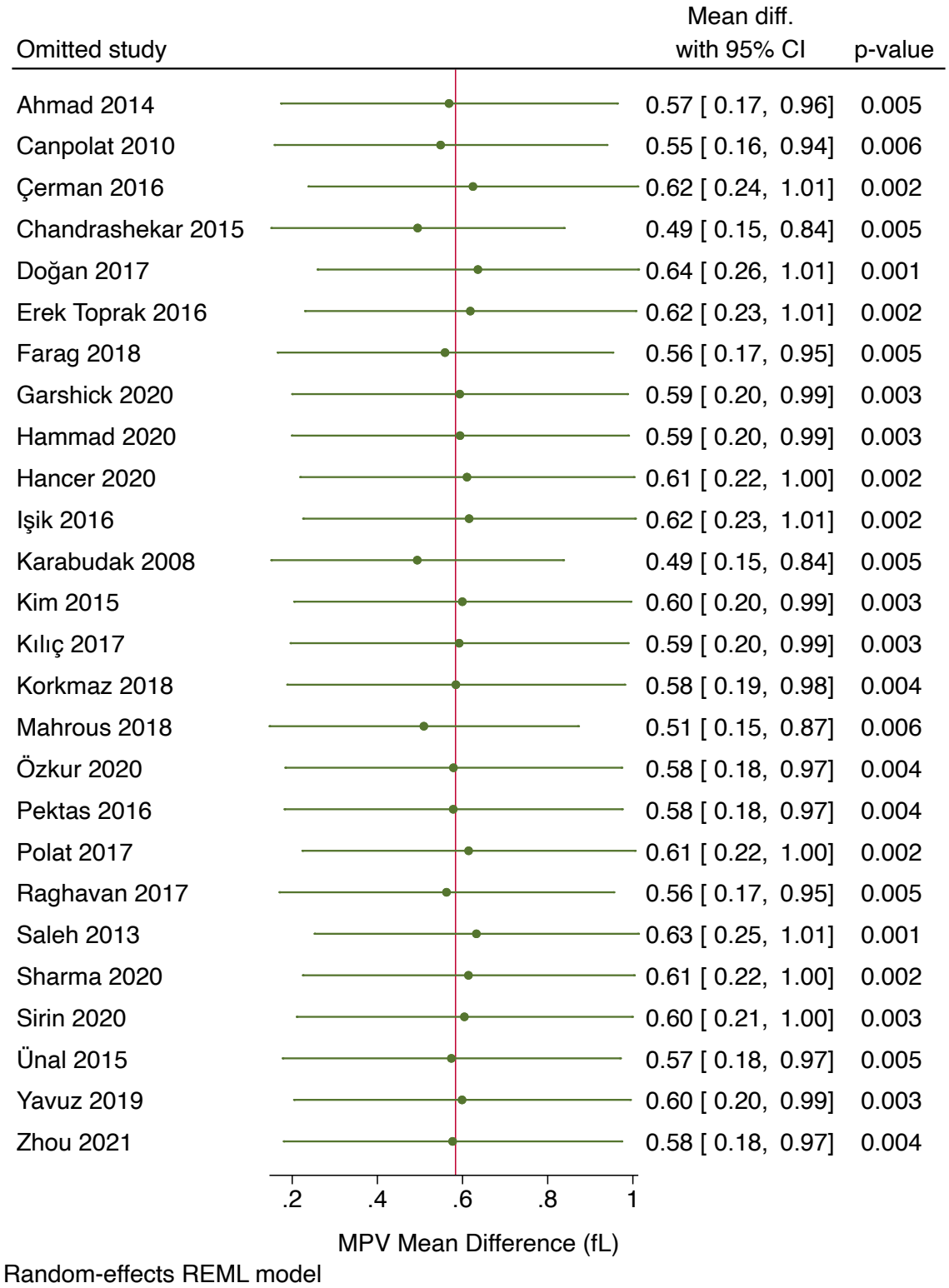

Online resource 3: Leave-one-out sensitivity analysis for PLT and psoriasis presence.

## Online Resource 6



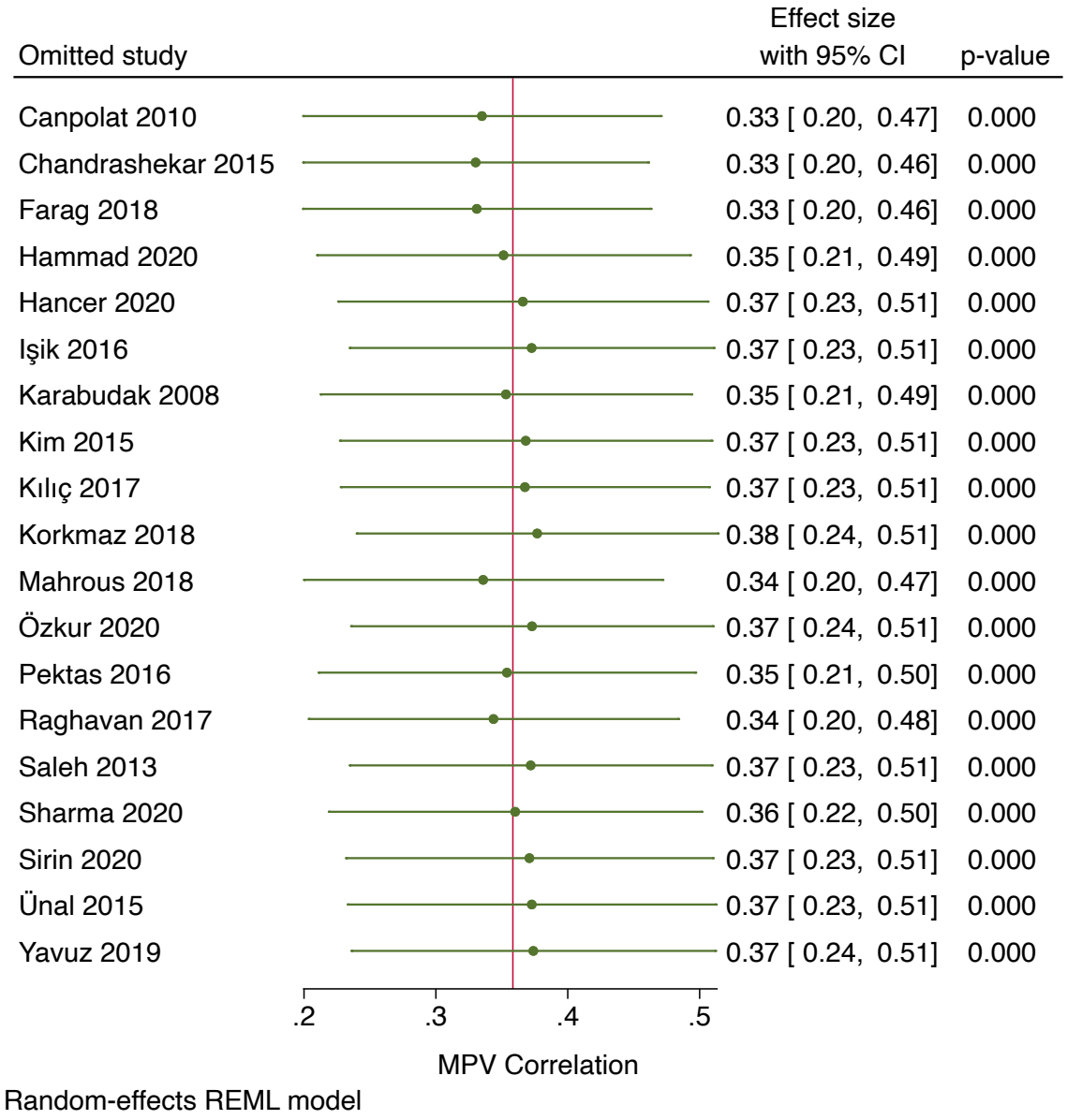

Online resource 6: Leave-one-out sensitivity analysis for PLT and psoriasis severity.

## Online Resource 7



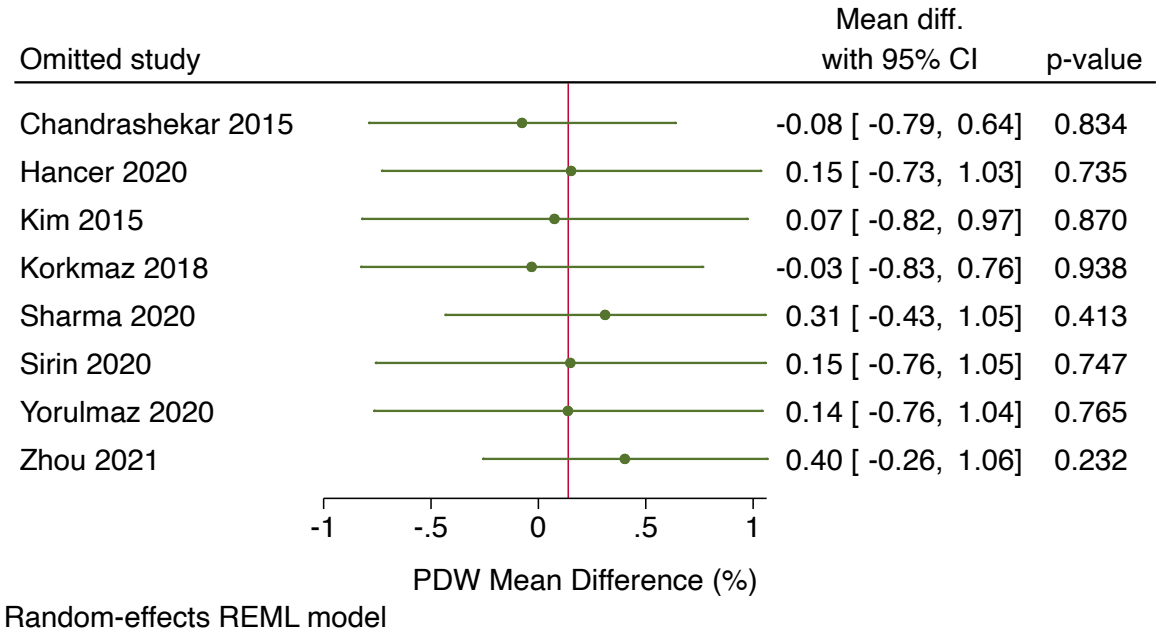

Online resource 7: Leave-one-out sensitivity analysis for MPV and psoriasis presence.

## Online Resource 10



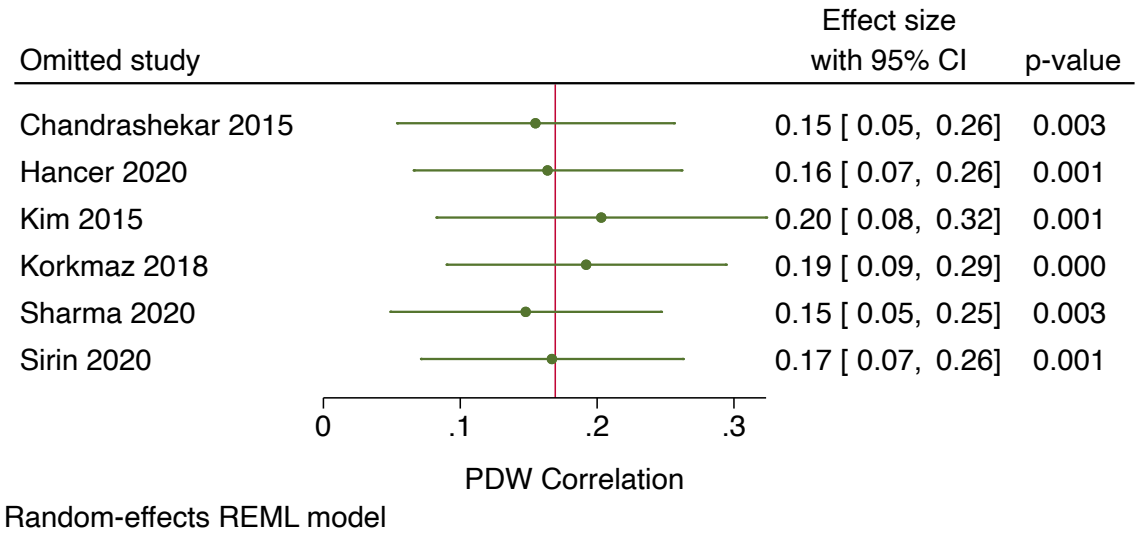

Online resource 10: Leave-one-out sensitivity analysis for MPV and psoriasis severity.

## Online Resource 13




Online resource 13: Leave-one-out sensitivity analysis for PDW and psoriasis presence.

## Online Resource 14




Online resource 14: Leave-one-out sensitivity analysis for PDW and psoriasis severity.

# Meta-regression Tables

## Online Resource 4: Meta-regression results of PLT and psoriasis

| Covariate | k | Regression coefficient – magnitude of effect size modification | p-value | 95% CI (lb, ub) | Heterogeneity accounted for | Residual Heterogeneity ($I^2$) |
|---|---|---|---|---|---|---|
| Smoking Prevalence | 10 | 62.9691 | 0.0150 | 12.2312 113.7069 | 100.00% | 0.00% |
| Cardiovascular Disease Prevalence | 14 | 733.8575 | 0.0199 | 116.1744 1351.5406 | 61.83% | 19.19% |
| Dyslipidaemia Prevalence | 13 | 61.8588 | 0.0219 | 8.9671 114.7505 | 61.23% | 23.19% |
| Hypertension Prevalence | 16 | 165.5206 | 0.0403 | 7.3242 323.7169 | 57.09% | 19.57% |
| Psoriatic Arthritis Prevalence | 13 | -45.0166 | 0.0868 | -96.5309 6.4976 | 35.61% | 26.97% |
| Diabetes Prevalence | 17 | 442.3163 | 0.1045 | -91.7623 976.3949 | 9.63% | 71.20% |
| Study Design: Prospective vs retrospective (reference) | 24 | 9.4656 | 0.2012 | -5.0485 23.9797 | 3.81% | 68.70% |
| Duration of Psoriasis | 11 | -0.9929 | 0.4827 | -3.7651 1.7792 | 0.00% | 43.99% |
| Percentage male | 24 | -17.3707 | 0.6422 | -90.6396 55.8982 | 0.00% | 69.54% |
| PASI Score | 16 | 0.2541 | 0.7781 | -1.5133 2.0214 | 0.00% | 67.75% |
| Mean age | 24 | -0.1233 | 0.8900 | -1.8710 1.6243 | 0.00% | 70.31% |
| Sample Size | 24 | 0.0010 | 0.9177 | -0.0182 0.0202 | 0.00% | 69.23% |
| Number of study centres | 24 | All studies single centre | | | | |

## Online Resource 8: Meta-regression results of MPV and psoriasis

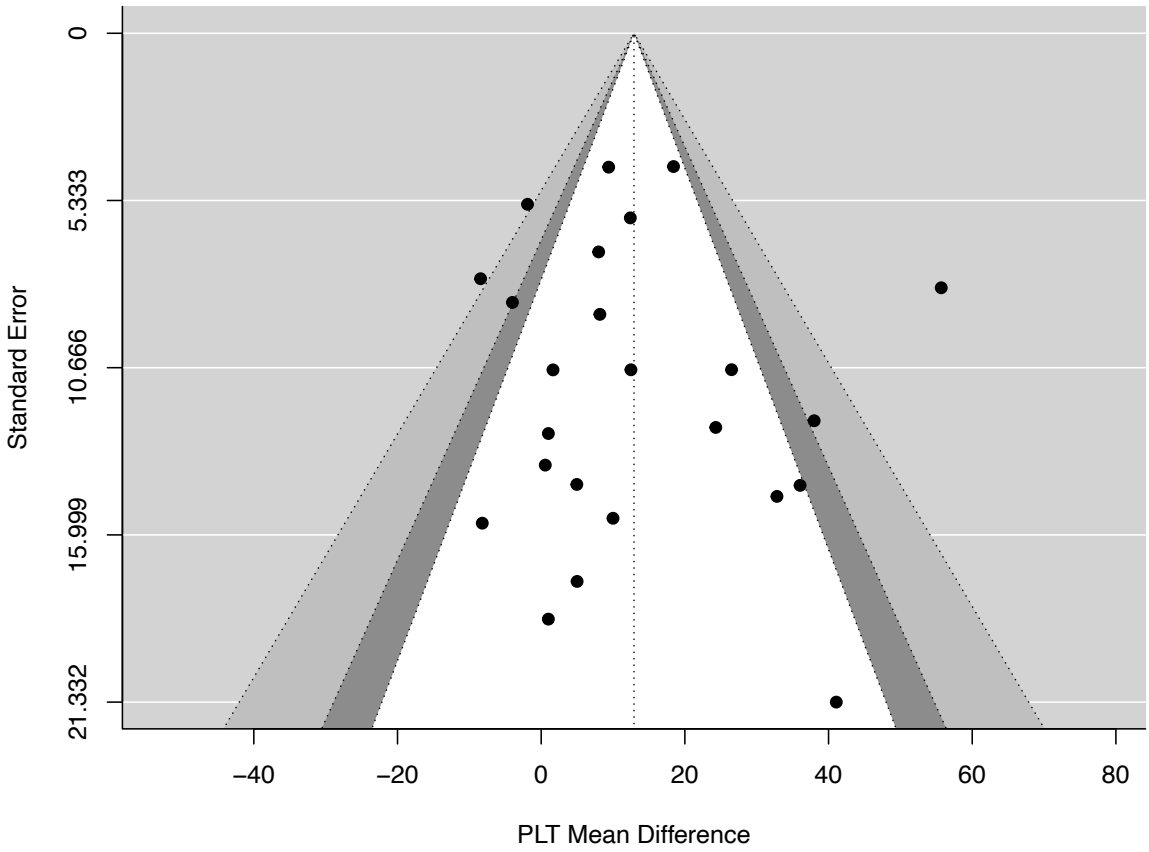
| Covariate | k | Regression coefficient – magnitude of effect size modification | p-value | 95% CI (lb, ub) | Heterogeneity accounted for | Residual Heterogeneity ($I^2$) |
|---|---|---|---|---|---|---|
| Percentage male | 26 | 5.3259 | <0.0001 | 3.4473 7.2044 | 60.49% | 91.24% |
| Duration of Psoriasis | 17 | -0.1430 | 0.0082 | -0.2489 - 0.0371 | 29.32% | 95.43% |
| Diabetes Prevalence | 21 | -16.0551 | 0.2558 | -43.7435 11.6333 | 1.25% | 96.67% |
| Psoriatic Arthritis Prevalence | 10 | 0.6787 | 0.4310 | -1.0106 2.3680 | 0.00% | 88.13% |
| Hypertension | 19 | -5.5196 | 0.4928 | -21.2918 10.2526 | 0.00% | 96.27% |
| Study Design: Prospective vs retrospective (reference) | 26 | 0.2114 | 0.5888 | -0.5550 0.9778 | 0.00% | 96.28% |
| Mean age | 26 | -0.0194 | 0.5911 | -0.0901 0.0513 | 0.00% | 96.44% |
| Smoking | 12 | 0.9311 | 0.7022 | -3.8412 5.7034 | 0.00% | 96.34% |
| Cardiovascular Disease Prevalence | 18 | -14.9378 | 0.7028 | -87.3219 58.8630 | 0.00% | 97.68% |
| Sample Size | 26 | 0.0001 | 0.9491 | -0.0019 0.0020 | 0.00% | 96.10% |
| PASI Score | 19 | -0.0005 | 0.9902 | -0.0848 0.0838 | 0.00% | 96.14% |
| Number of study centres | 26 | All studies single centre | | | | |
| Dyslipidaemia | 16 | Dyslipidaemia an exclusion criteria in all studies | | | | |

## Online Resource 11: Meta-regression results of MPV and PASI

| Covariate | k | Regression coefficient – magnitude of effect size modification | p-value | 95% CI (lb, ub) | Heterogeneity accounted for | Residual Heterogeneity ($I^2$) |
|---|---|---|---|---|---|---|
| Percentage male | 19 | 1.1194 | 0.0037 | 0.3645 1.8742 | 33.45% | 88.10% |
| Psoriatic Arthritis Prevalence | 7 | 0.7192 | 0.0940 | -0.1226 1.5611 | 33.25% | 78.56% |
| Duration of Psoriasis | 14 | -0.0232 | 0.2223 | -0.0605 0.0141 | 4.49% | 92.41% |
| Smoking Prevalence | 11 | 0.7563 | 0.2339 | -0.4890 2.0016 | 5.45% | 89.91% |
| Study Design: Prospective vs retrospective (reference) | 19 | -0.1596 | 0.2381 | -0.4248 0.1056 | 3.63% | 91.50% |
| PASI Score | 16 | 0.0141 | 0.2876 | -0.0119 0.0401 | 2.48% | 92.70% |
| Diabetes Prevalence | 17 | -2.9177 | 0.5393 | -12.2326 6.3973 | 0.00% | 93.76% |
| Mean age | 19 | 0.0069 | 0.5629 | -0.0165 0.0304 | 0.00% | 92.43% |
| Cardiovascular Disease Prevalence | 16 | -6.3091 | 0.5868 | -29.0638 16.4456 | 0.00% | 94.35% |
| Hypertension Prevalence | 15 | -2.2887 | 0.5938 | -10.6992 6.1219 | 0.00% | 92.08% |
| Sample Size | 19 | -0.0003 | 0.6495 | -0.0014 0.0009 | 0.00% | 92.31% |
| Number of study centres | 18 | All studies single centre | | | | |
| Dyslipidaemia Prevalence | 13 | Dyslipidaemia an exclusion criteria in all studies | | | | |

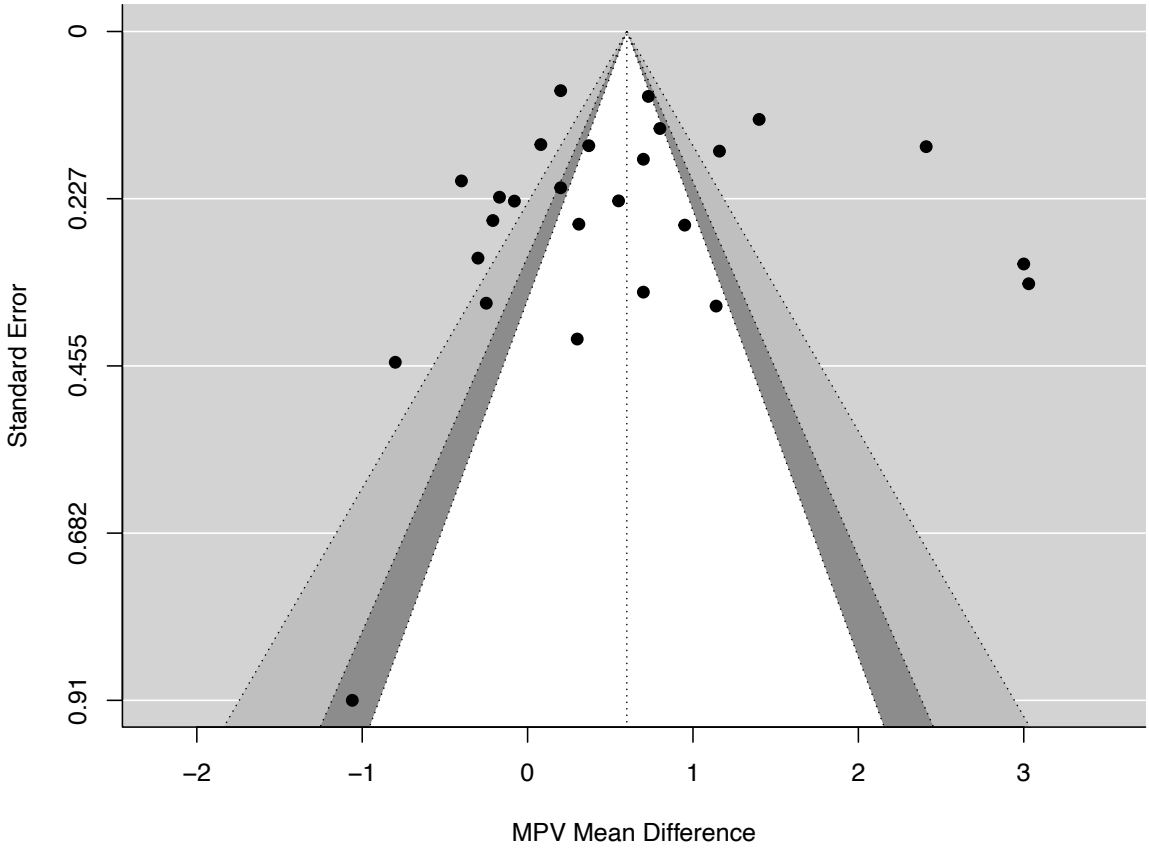
# Supplemental Figures

## Online Resource 5



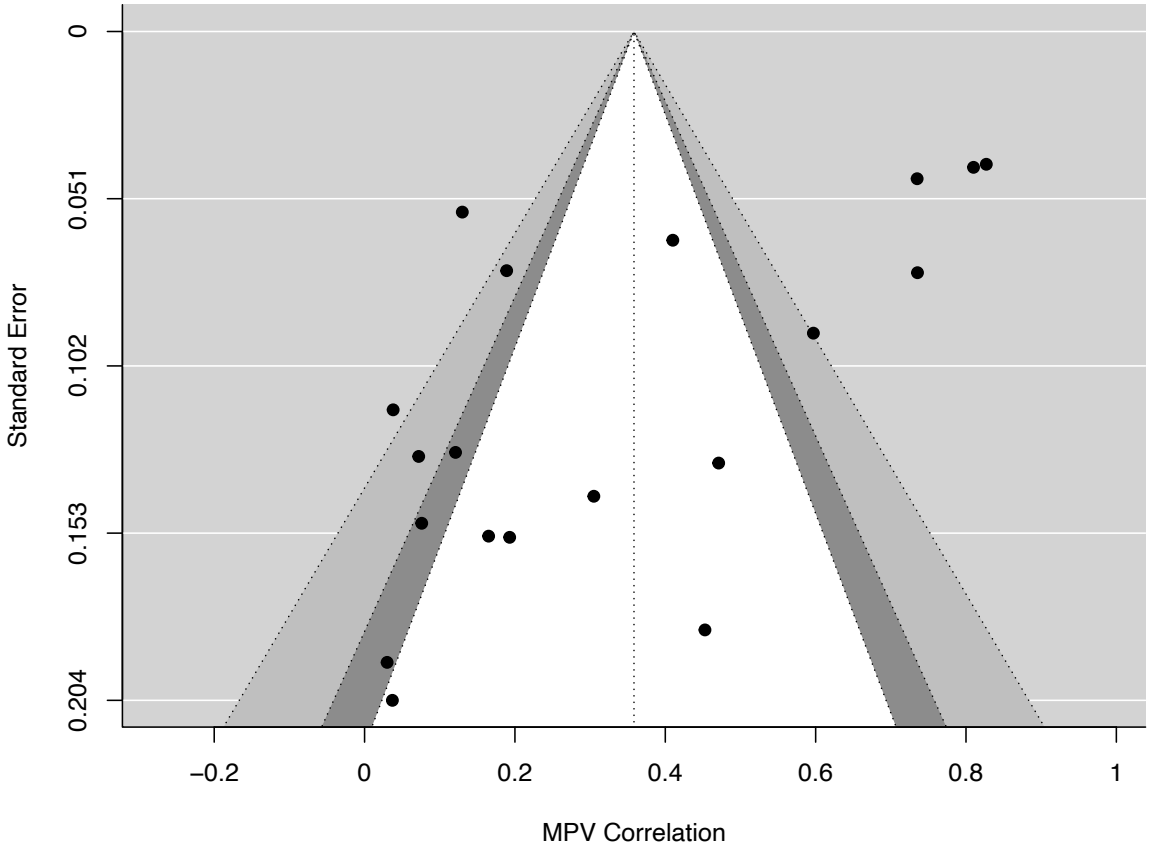

Online Resource 5: Contour enhanced funnel plot for PLT and psoriasis.

## Online Resource 9




Online Resource 9: Contour enhanced funnel plot for MPV and psoriasis.

## Online Resource 12




Online Resource 12: Contour enhanced funnel plot for MPV and PASI.